# SINGLE LINE DIAGRAM - 415 V LT DISTRIBUTION WITH DG CHANGEOVER

(REFERENCE DRAWING - ALL VALUES ASSUMED, SEE NOTES)

**DRAWN BY SUVADEEP MALLIK**

### LOAD SUMMARY (ASSUMED)

CONNECTED LOAD : 46.5 kW

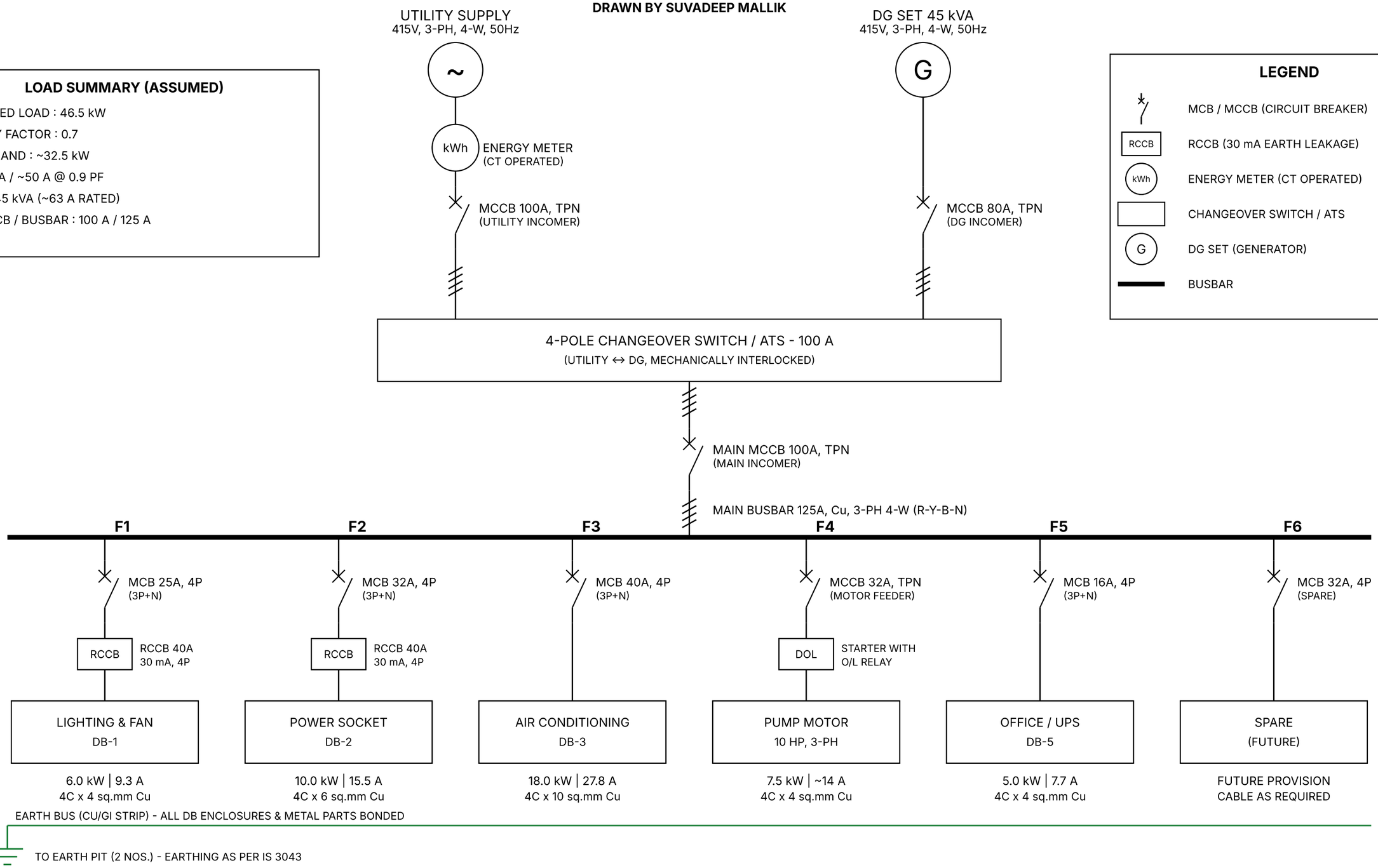
DIVERSITY FACTOR : 0.7

MAX. DEMAND : ~32.5 kW

= ~36 kVA / ~50 A @ 0.9 PF

DG SET : 45 kVA (~63 A RATED)

MAIN MCCB / BUSBAR : 100 A / 125 A




### LEGEND

- MCB / MCCB (CIRCUIT BREAKER)
- RCCB — RCCB (30 mA EARTH LEAKAGE)
- kWh — ENERGY METER (CT OPERATED)
- CHANGEOVER SWITCH / ATS
- G — DG SET (GENERATOR)
- BUSBAR

### NOTES:

1. ALL RATINGS, LOADS AND CABLE SIZES ARE ASSUMED FOR A TYPICAL SMALL COMMERCIAL LOAD - REPLACE WITH ACTUAL VALUES.
2. CABLES: 4-CORE Cu XLPE ARMOURED, INDICATIVE SIZES (DERATING / VOLTAGE DROP / FAULT LEVEL NOT CHECKED).
3. UTILITY AND DG MUST NEVER BE PARALLELED - CHANGEOVER IS INTERLOCKED. N-E BONDING PER LOCAL CODE.
4. RCCB 30 mA PROVIDED ON LIGHTING AND SOCKET CIRCUITS FOR PERSONNEL PROTECTION.
5. REFERENCE DRAWING ONLY - VERIFY WITH A LICENSED ELECTRICAL ENGINEER BEFORE INSTALLATION.

## SINGLE LINE DIAGRAM

415 V LT DISTRIBUTION WITH DG CHANGEOVER

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|---|---|
| DATE: 05-11-2026 | SCALE: NTS |
| DRAWN BY: SUVADEEP MALLIK | CHECKED BY: _____ |
| SHEET: 1 OF 1 | REV: A |